

RESEARCH REPORT

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Clinical factors, diagnostic delay, and residual deficits in chronic inflammatory demyelinating polyradiculoneuropathy

Carina Bunschoten¹  | Patricia H. Blomkwist-Markens² | Anja Horemans² | Pieter A. van Doorn¹ | Bart C. Jacobs^{1,3}

¹Department of Neurology, Erasmus MC, University Medical Center Rotterdam, Rotterdam, The Netherlands

²National Patient Organization for Neuromuscular Diseases, Baarn, The Netherlands

³Department of Immunology, Erasmus MC, University Medical Center Rotterdam, Rotterdam, The Netherlands

Correspondence

Carina Bunschoten, MD, Department of Neurology, Erasmus MC, University Medical Center Rotterdam, Wytemaweg 80, 3015 CN Rotterdam, The Netherlands.
Email: c.bunschoten@erasmusmc.nl

Abstract

Management of patients with chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) is complicated by the challenging diagnosis, monitoring of disease activity, and treatment response. Real world data about the current practice of care in CIDP are rare, but important to improve clinical guidelines. In this study, we determined the current practice of diagnosis and treatment in relation to the clinical outcome of patients with CIDP in a cross-sectional study in The Netherlands. All patients registered at the Dutch neuromuscular patient organization and patients at the Erasmus Medical Center were approached. CIDP diagnosis was confirmed by using patient files and expert consensus. The response rate was 137/197 (70%) and the diagnosis CIDP was confirmed in 112 patients. The time from onset of symptoms until CIDP diagnosis was median 5 months and > 12 months in 26% of the patients. Patients with a late diagnosis (>5 months) compared to patients with an early diagnosis (≤ 5 months) more frequently had another initial diagnosis ($P < .001$), multifocal abnormalities ($P = .011$), final diagnosis made in university hospitals ($P = .001$), and more (residual) complaints, also illustrated via two patient reported, validated clinical outcome measures. Although 97% of patients received treatment, 88% reported (residual) neurological symptoms and deficits. CIDP diagnosis is often delayed, especially in patients with atypical CIDP variants. Diagnostic delay may result in delayed initiation of treatment. A more rapid and accurate diagnosis of CIDP may prevent or at least reduce residual neurological deficits and accompanying disability in CIDP.

KEYWORDS

chronic inflammatory demyelinating polyradiculoneuropathy, clinical outcome measures, diagnosis, residual neurological deficits, treatment

1 | INTRODUCTION

Chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) is typically characterized by progressive symmetrical weakness and sensory deficits in the legs and arms. However, CIDP may have a range of clinical presentations due to the existence of atypical CIDP variants including asymmetric, focal, and variants with predominant distal, sensory, or motor involvement.¹⁻⁴ At least 15 sets of diagnostic criteria for CIDP have been developed. The most frequently used set in clinical

practice and research are the European Federation of Neurological Societies/Peripheral Nerve Society 2010 diagnostic criteria for CIDP (EFNS/PNS 2010).^{1,5-7} The different diagnostic criteria, together with the infrequency, heterogeneity, and extensive differential diagnosis of chronic neuropathies, make the diagnosis of CIDP challenging.⁸⁻¹⁴ CIDP is one of the few treatable neuropathies that usually responds to treatment with either intravenous immunoglobulins (IVIg), corticosteroids, or plasma exchange (PE).¹⁵⁻¹⁸ If one treatment fails, one of the other treatment types is often effective.¹⁹ The treatment of CIDP is complicated

by the variable responsiveness of individual patients. Although dosing algorithms for IVIg treatment are suggested, there is no gold standard available on how to select treatment type, regimen, and how to monitor maintenance treatments.^{20,21} Early diagnosis and optimal treatment are important to prevent secondary axonal nerve damage which can cause permanent disability in patients.^{22,23} Treatment responsiveness varies among patients, this may lead to under treatment, as well as over-treatment when the disease is in remission, as shown in two randomized controlled studies where 16% to 55% of placebo treated CIDP patients did not relapse after discontinuation of treatment.²⁴⁻²⁹

There are limited real world data available on the current practice of CIDP diagnosis and treatment, and disease impact for patients and society. Here, we conducted a cross-sectional study in patients with CIDP in The Netherlands, to determine the current practice of CIDP diagnosis and treatment.

2 | MATERIALS AND METHODS

2.1 | Patients

For this cross-sectional study, we approached all members registered with the combined GBS/CIDP diagnosis group at the Dutch national patient organization for neuromuscular diseases ("Spierziekten Nederland") and the patients referred to the Erasmus MC between 2010 and 2015. Only patients, in whom the CIDP diagnosis and treatment response could be verified, were included in this study. We used patients' hospital files and the EFNS/PNS 2010 criteria to verify the diagnosis.¹ When not all diagnostic information was available (such as limited nerve conduction results), an expert consensus on the diagnosis (CIDP or other disease) was made by three investigators (CB, researcher, and two experienced neuromuscular experts PvD and BJ). Patients were included independent of age, patients' history, disease severity, or treatment. The study was approved by the ethical committee of the Erasmus Medical Center in Rotterdam. All patients gave written informed consent for participation in the study, and for obtaining discharge and outpatient clinic letters.

2.2 | Data questionnaire and collection

Data were collected according to a predefined protocol and questionnaire regarding demography (age, gender, and hospital), medical history, the timing and work-up of diagnosis, treatment, follow-up, current symptoms and disability. Furthermore two standardized and validated clinical outcome measures for immune-mediated neuropathies were assessed: (a) the Rasch-modified Fatigue Severity Scale (R-FSS)³⁰ which is used to capture the consequences of fatigue with scores ranging from 0 ("no signs of fatigue") to 21 ("most disabling fatigue"). (b) the Rasch-built Overall Disability Scale (R-ODS)³¹ which is used to capture activity and social participation limitations, with scores ranging from 0 ("most severe limitations") to 48 ("no limitations"). The CIDP Disease Activity Status (CDAS)²⁷ is a simple tool to classify CIDP patients according to disease activity and treatment status. CDAS scores range from 1. Cured ("Stable disease, off treatment ≥ 5 years") to 5. Unstable active disease ("Treatment naïve" or "progressive/relapsing disease course, on/off treatment").

2.3 | Data and statistical analysis

Data were analyzed using SPSS statistics 24. To investigate the influence of the time between onset of symptoms and the diagnosis of CIDP, patients were divided in two groups based on the median value, labeled as "Early diagnosis" and "Late diagnosis." Continuous variables were described using median values and interquartile ranges (IQRs). Categorical data were described using frequencies and percentages. Mann-Whitney *U* test was used to compare medians, and Fisher exact test or χ^2 to compare proportions. Kaplan-Meier analysis was used to illustrate the timing of diagnosis and the Log-Rank test was used to compare the curves of time to diagnosis. Two-sided *P*-values of $<.05$ were considered to be statistically significant.

3 | RESULTS

Of the 197 patients approached, 137 (70%) patients gave approval to participate in the study. Twenty-five patients (19%) were excluded from the analysis for various reasons, including 18 patients from the GBS/CIDP

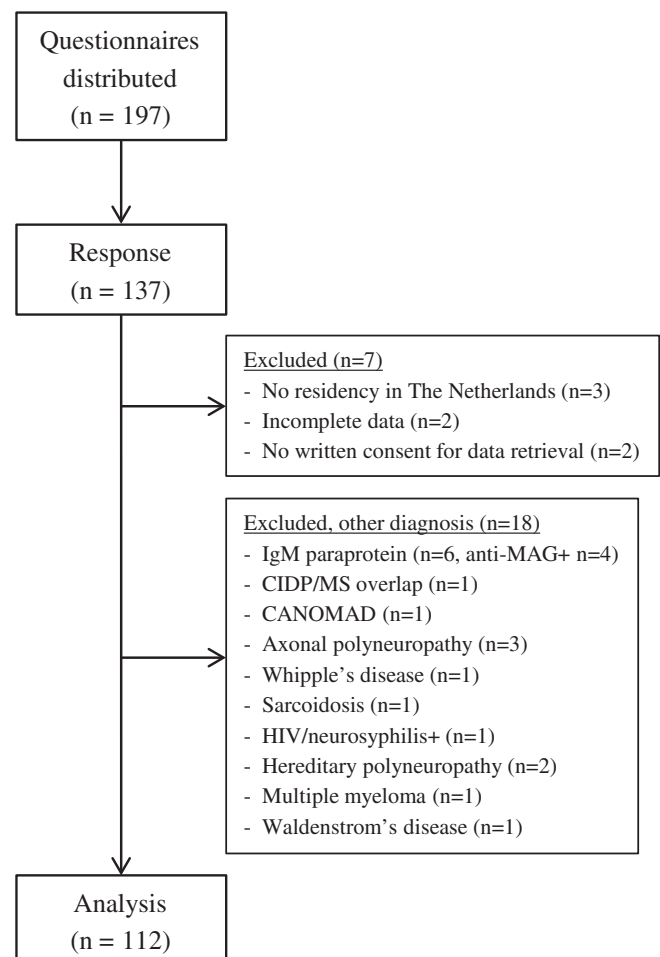


FIGURE 1 Patient inclusion. Abbreviations: CANOMAD, chronic ataxic neuropathy, ophthalmoplegia, monoclonal IgM protein, cold agglutinins and disialosyl antibodies; HIV, human immunodeficiency virus; IgM, immunoglobulin-M; MAG, myelin-associated glycoprotein; MS, multiple sclerosis

diagnosis group with known diagnosis other than CIDP (Figure 1). Diagnostic consensus was reached in the 112 remaining patients. Fifteen of these patients have been described previously.¹⁹

3.1 | Clinical characteristics and diagnostic work-up

The patients originated from all provinces in The Netherlands. Demographical, clinical, and diagnostic characteristics are summarized in Table 1. At study entry, patients had a median age of 62 years (IQR 54-69 years). The time between the onset of symptoms related to CIDP (including muscle weakness, evident sensory deficits, or pain) and correct CIDP diagnosis was highly variable, with a median duration of 5 months (IQR 2-12 months, Figure 2).

In 29 (26%) patients the time to CIDP diagnosis was more than 1 year.

3.2 | Comparison early and late diagnosis of CIDP

Patients were divided in two groups, based on time to diagnosis within or after the median duration of 5 months (labeled early and late diagnosis, respectively). Patients with a late diagnosis compared to an early diagnosis more frequently were diagnosed with an initial other disease (31% vs 2%, $P < .001$), including hereditary polyneuropathies ($n = 5$), chronic idiopathic axonal polyneuropathy ($n = 1$), IgM paraprotein related neuropathy ($n = 1$), IgG paraprotein related neuropathy ($n = 1$), myasthenia gravis ($n = 1$), bulging disc/tendomyalgia

TABLE 1 Demographics and diagnostic characteristics

	All patients (n = 112)	Early diagnosis (n = 61)	Late diagnosis (n = 51)	P-value
<i>General characteristics</i>				
Sex (male), n (%)	72 (64)	37 (61)	35 (69)	ns
Age at study entry (years), median (IQR)	62 (54-69)	61 (50-69)	64 (57-70)	ns
Age at onset (years), median (IQR)	52 (40-61)	54 (42-62)	50 (40-60)	ns
<i>Clinical and diagnostic features</i>				
Time from onset to CIDP diagnosis (months), median (IQR)	5 (2-12)	3 (2-3)	12 (8-24)	n.a.
CIDP disease duration (years), median (IQR)	9 (6-15)	7 (5-11)	13 (8-20)	<.001
<i>Presenting symptoms, n (%)</i>				
Muscle weakness	62 (55)	39 (64)	23 (45)	.046
Sensory symptoms	89 (80)	51 (84)	38 (75)	ns
Pain	14 (13)	8 (13)	6 (12)	ns
Headache, cranial nerve deficits	1 (1)	-	1 (2)	ns
<i>Initial diagnosis, n (%)</i>				
CIDP	64 (57)	35 (57)	29 (57)	ns
GBS or A-CIDP	31 (28)	25 (41)	6 (12)	n.a.
Other ^a	17 (15)	1 (2)	16 (31)	<.001
<i>CIDP phenotypes, n (%)</i>				
Typical CIDP	90 (80)	52 (85)	38 (75)	ns
Atypical CIDP variants	22 (20)	9 (15)	13 (25)	ns
Asymmetric	9 (8)	1 (2)	8 (16)	.011
Predominant motor	7 (6)	3 (5)	4 (8)	ns
Predominant sensory	4 (4)	4 (7)	-	ns
Predominant distal	2 (2)	1 (2)	1 (2)	ns
<i>Hospital, n (%)</i>				
First visit in university hospital	17 (15)	10 (16)	7 (14)	ns
CIDP diagnosis in university hospital	77 (69)	34 (56)	43 (84)	.001
Lumbar puncture performed, n (%)	96/105 (91)	58/58 (100)	38/47 (81)	<.001
Protein level elevated, n (%)	71/95 (75)	44/57 (77)	27/38 (71)	ns
Cell count, median (IQR)	2 (1-5) (n = 80)	2 (1-6) (n = 80)	2 (1-4) (n = 80)	ns

Note: Patients were dichotomized based on the time until CIDP diagnosis as follows: Early diagnosis (≤ 5 months) and Late diagnosis (> 5 months). Abbreviations: (A-)CIDP, (acute-onset) chronic inflammatory demyelinating polyradiculoneuropathy; GBS, Guillain-Barré syndrome; IQR, interquartile range; n.a., not applicable; ns, not significant.

^aOther diagnoses ($n = 17$): hereditary polyneuropathies ($n = 5$), chronic idiopathic axonal polyneuropathy ($n = 1$), IgM paraprotein related neuropathy ($n = 1$), IgG paraprotein related neuropathy ($n = 1$), myasthenia gravis ($n = 1$), bulging disc/tendomyalgia ($n = 4$), motor neuron disease ($n = 2$), paraneoplastic syndrome ($n = 1$), polyneuropathy due to kidney dysfunction ($n = 1$).

($n = 4$), motor neuron disease ($n = 2$), and paraneoplastic syndrome ($n = 1$). Patients with a late diagnosis also more frequently had atypical CIDP variants, mainly asymmetrical forms (also labeled multifocal acquired demyelinating sensory and motor neuropathy (MADSAM) or Lewis-Sumner Syndrome [LSS]; $P = .011$, Table 1). Furthermore, 84% of patients with a late diagnosis were finally diagnosed in a university hospital compared to 56% in the early diagnosis group ($P = .001$).

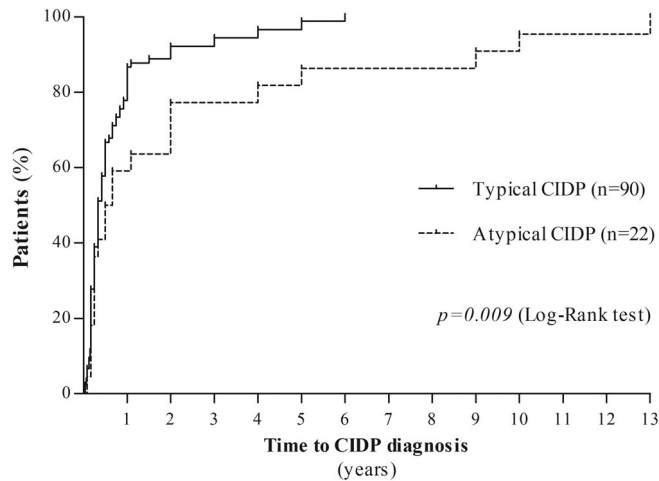


FIGURE 2 Time between onset of symptoms and diagnosis in typical chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) and atypical CIDP variants

Similar associations were found when the patients were divided in two groups based on 12 months to CIDP diagnosis (*data not shown*) or divided based on the clinical phenotype (typical CIDP vs atypical CIDP variants). The time to CIDP diagnosis was longer in patients with atypical CIDP variants than in patients with typical CIDP (7 vs 4 months, Log-Rank test $P = .009$, Figure 2).

3.3 | Treatment

An overview of the previous and current treatments, and the treatment response is provided in Table 2. Induction treatment was started in 109 patients after whom 79% initially improvement. Three patients were never treated because of mild symptoms or because they refused treatment.

3.4 | Current clinical state

The fast majority of patients (88%) reported current (residual) neurological symptoms related to CIDP despite treatment, including muscle weakness (73%), sensory symptoms (77%), pain (52%), and fatigue (77%; Table 3). Classification according to the CIDP Disease Activity Status (CDAS) showed that the late diagnosis group was more often classified as "3. Stable active disease, on treatment" (73% vs 46%, $P = .004$). In contrast, the early diagnosis group were more frequently classified as "Remission" (26% vs 10%, $P = .027$). Patients with a late diagnosis reported both more signs of fatigue on the R-FSS and more

TABLE 2 Treatment strategies and treatment response

	Induction treatment ($n = 112$)	Maintenance treatment ^a ($n = 112$)		
IVIg, n (%)	90 (80)	58 (52)		
Corticosteroids, n (%)	13 (12)	8 (7)		
IVIg + corticosteroids, n (%)	6 (5)	3 (3)		
Plasma exchange, n (%)	-	2 (2)		
Prednisone + methotrexate, n (%)	-	1 (1)		
Not treated, n (%)	3 (3)	40 (36)		
	All patients ($n = 112$)	Early diagnosis ($n = 61$)	Late diagnosis ($n = 51$)	<i>P</i> -value
Currently treated, n (%)	72 (64)	35 (57)	37 (73)	ns
Treatment duration ^b (years), median (IQR)				
Previously treated patients	1.8 (0.4-4.0)	1.9 (0.4-4.0)	1.5 (0.3-4.8)	ns
Currently treated patients	7.0 (3.1-10.4)	4.5 (2.3-8.3)	8.8 (4.0-15.5)	.009
Never treated, n (%)	3 (3)	2 (3)	1 (2)	ns
Response to induction treatment ($n = 109$), n (%)				
Clinical recovery	9 (8)	6 (10)	3 (6)	ns
Clinical improvement but further treatment needed	65 (58)	32 (53)	33 (65)	ns
Clinical improvement but insufficient to continue treatment	12 (11)	7 (12)	5 (10)	ns
No clinical improvement	23 (20)	14 (23)	9 (18)	ns

Note: Patients were dichotomized based on the time until CIDP diagnosis as follows: Early diagnosis (≤ 5 months) and Late diagnosis (> 5 months).

Abbreviations: IQR, interquartile range; IVIg, intravenous immunoglobulins; ns, not significant.

^aNumber of patients on maintenance treatment at time of the study entry.

^bTotal duration of treatments given from disease onset until study entry.

TABLE 3 Current clinical state

	All patients (n = 112)	Early diagnosis (n = 61)	Late diagnosis (n = 51)	P-value
<i>Current symptoms</i>				
Current neurological symptoms, n (%)	98 (88)	54 (89)	44 (86)	ns
Muscle weakness	82 (73)	40 (66)	42 (82)	.046
Sensory symptoms	86 (77)	47 (77)	39 (77)	ns
Pain	58/111 (52)	31/60 (52)	27 (53)	ns
Fatigue	86 (77)	46 (75)	40 (78)	ns
Fatigue on R-FSS (n = 86), median (IQR)	17 (14-20)	15 (13-19)	18 (16-21)	.036
Disability on R-ODS (n = 107), median (IQR)	37 (30-44)	39 (35-46)	34 (27-42)	.008
Additional support ^a , n (%)	49/111 (44)	26/60 (43)	23 (45)	ns
<i>CIDP disease activity status, n (%)</i>				
1. Cured	17 (15)	10 (16)	7 (14)	ns
2. Remission	21 (19)	16 (26)	5 (10)	.027
3. Stable active disease	65 (58)	28 (46)	37 (73)	.004
4. Improving	5 (4)	5 (8)	-	ns
5. Unstable active disease	4 (4)	2 (3)	2 (4)	ns

Note: Patients were dichotomized based on the time until CIDP diagnosis as follows: Early diagnosis (≤ 5 months) and Late diagnosis (> 5 months). Abbreviations: IQR, interquartile range; ns, not significant; R-FSS, Rasch-built Fatigue Severity Scale; R-ODS, Rasch-built Overall Disability Scale.

^aIncluding physical therapy, rehabilitation, home care, informal care, and other (psychologist, hydro-therapist, posture, and movement education).

limitations and disability on the R-ODS ($P = .03$ and $P = .008$, respectively; Table 3, Figure 3).

4 | DISCUSSION

Our study confirms the complexity of diagnosis and the management of CIDP in current clinical practice. First, a substantial proportion of patients with CIDP in this study were diagnosed with an initial other disease. While previous studies indicated that patients diagnosed with CIDP in up to 54% of patients may actually have another diagnosis^{10,12,32,33} (CIDP overdiagnosis), our study suggest that CIDP underdiagnosis also occurs. CIDP underdiagnosis appears to be mainly related to poor or limited performance and interpretation of clinical and diagnostic assessments. Atypical CIDP variants are previously reported as a pitfall for CIDP overdiagnosis, but our study shows the opposite might also be true, further underlining the challenging diagnosis of atypical CIDP variants. Second, the time between onset of symptoms and diagnosis of CIDP was 5 months and in one-quarter of patients more than 1 year. In the late diagnosis group, the atypical CIDP variants (especially the asymmetric variant), were more prevalent. Most of these patients were finally diagnosed in university hospital, which is in line with a previous proposal from neuromuscular experts, concerning the consultation of specialized neuromuscular centers in atypical cases of persisting diagnostic uncertainty.¹¹ Third, the large majority of patients reported (residual) neurological symptoms or deficits at time of the study, often despite long-term treatment.

CIDP is regarded as a disorder with chronic or relapsing-remitting inflammation of peripheral nerves that requires immune-modulating

treatment during active disease to prevent secondary axonal nerve damage and related permanent disability.³⁴ The current study indicates that a later diagnosis may be related to a more protracted disease course requiring prolonged treatment, and more extensive residual neurological complaints and disability. One could speculate whether these patients could have had a better clinical outcome if diagnosed and treated earlier, although this cannot be determined with the current study design. At study inclusion, patients with a late diagnosis have had CIDP for as long as median 13 years, compared to 7 years in the early diagnosis group. There is no evidence that the health care for CIDP has changed significantly during these years in The Netherlands.

Patients diagnosed later in the disease course are treated more often and for a longer period of time. One could speculate whether these patients still have active disease or are (over)treated for symptoms related to (secondary) axonal nerve damage rather than symptoms due to active nerve inflammation. Overtreatment in CIDP has been previously reported and needs to be avoided as much as possible as most treatments for CIDP are expensive.^{26,29} When patients receive maintenance treatment, stepwise tapering of the dose or extension of the interval is essential to determine disease activity in patients and prevent overtreatment. Algorithms to reduce IVIg doses according to the patients' individual needs are suggested.^{20,21}

A large number of patients in our study reported a range of (residual) neurological symptoms and deficits, which is further illustrated by the results on the standardized and validated clinical outcome measures R-ODS and R-FSS. Whether the reported neurological deficits can be explained by diagnostic delay and whether they could have been reduced or prevented by a more rapid diagnosis and subsequent early therapeutic options, is unclear.

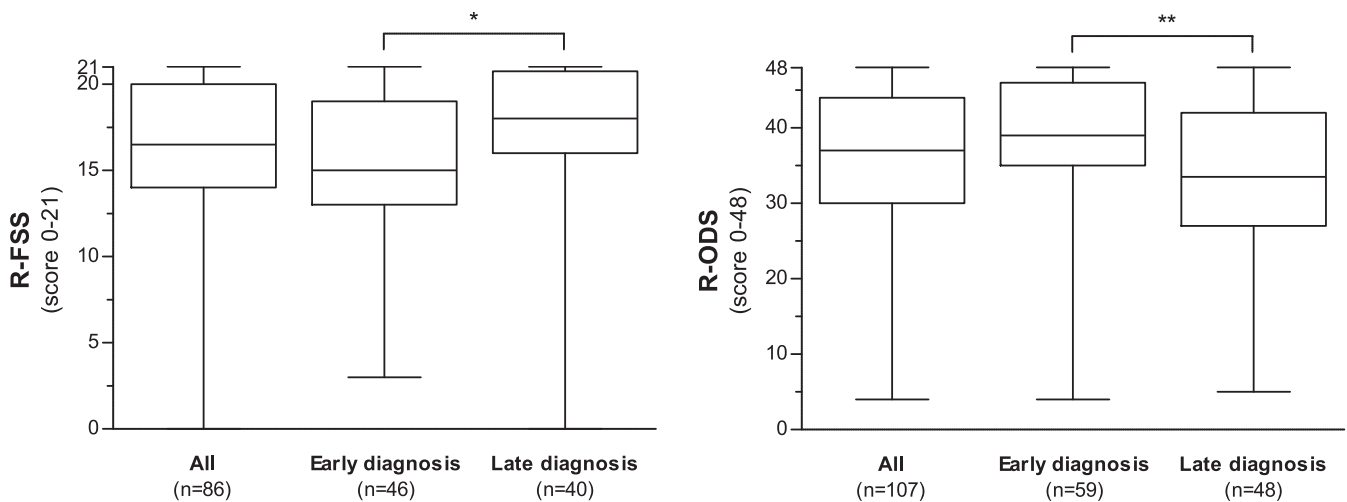


FIGURE 3 Current clinical state and timing of diagnosis. Current clinical state illustrated by the Rasch-built Fatigue Severity Scale (R-FSS; left) and the Rasch-built Overall Disability Scale (R-ODS; right). Patients were dichotomized based on the time until CIDP diagnosis as follows: Early diagnosis (≤ 5 months) and Late diagnosis (> 5 months). *Score on R-FSS $P = .036$, **Score on R-ODS $P = .008$

This study has several limitations. First, the study population was probably biased as patients with more active disease may be more motivated to enlist in a patient organization or are seen at the Erasmus MC as tertiary neuromuscular expertise hospital and Center of Excellence for CIDP. The exact frequency of CIDP in The Netherlands is unknown, but extrapolating a pooled prevalence rate of 2.81 per 100,000 persons (95% CI 1.58-4.39) from a recent systematic review and meta-analysis on CIDP,³⁵ the estimated number of CIDP patients would be around 472. This would imply that about one-third of all Dutch CIDP patients participated in this study. Second, the study lacks an age and sex matched healthy control groups for comparison of fatigue and disability scores. Third, the retrospective nature of this study could have limited the possibility to confirm the diagnosis on NCS (using the EFNS/PNS 2010 diagnostic criteria) and the timing of onset of symptoms, although we were able to extensively check these data in the hospital files. Furthermore, the study focused on the situation in the Netherlands that may differ from the management of CIDP elsewhere. All provinces of The Netherlands are represented in this cohort, but treatment preferences may vary between hospitals in The Netherlands. Our study thereby may over represent IVIg treated patients, as IVIg is the preferred first treatment in the Erasmus MC.

This study provides an overview of the diagnostic practice of CIDP, which shows a variation in the time to diagnosis, treatment, and outcome. To shorten the duration of the diagnostic work-up, especially in patients with an atypical CIDP variant, renewed upcoming CIDP guidelines on diagnosis and treatment from the European Academy of Neurology/Peripheral Nerve Society (EAN/PNS) together with results of ongoing CIDP treatment trials will be useful. Furthermore, the consultation of neuromuscular expertise centers should be considered early in the disease course, to hasten the diagnostic work-up, in order to make the correct diagnosis and to provide advice on a treatment plan. Prospective research with attention to (mis)diagnosis, the expanding CIDP spectrum (including the recently discovered

antibodies to [para]nodal proteins),³⁶ treatment, and outcome is important to further improve the current standard of diagnosis and care in CIDP patients.

CONFLICT OF INTEREST

P.B., C.B., and A.H. have none to disclose. P.vD. reports grants from Sanquin, grants from Prinses-Beatrix Spierfonds, grants from Baxalta, grants from Grifols, other from Octapharma, other from Kedrion, other from CSL-Behring, other from Hansa, outside the submitted work. B.J. reports grants from Baxalta, grants from Grifols, grants from CSL-Behring, grants from Annexon, grants from Prinses-Beatrix Spierfonds, grants from GBS-CIDP Foundation International, outside the submitted work.

ORCID

Carina Bunschoten  <https://orcid.org/0000-0002-6116-5656>

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